

Unit 3: Knowledge Representation and Probabilistic Reasoning

SubUnit 3.2: Knowledge Representation Techniques and Issues

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Knowledge Representation: Overview

What is Knowledge Representation?

The method of encoding information about the world in a form that a computer system can utilize to solve complex tasks.

Why is KR Important?

- Enables machines to reason and make decisions
- Bridges human knowledge and machine processing
- Foundation for intelligent behavior

Today's Focus:

- Major KR techniques
- Key issues and trade-offs in representation

Reference: Russell & Norvig (2020), Chapter 8-9; Rich et al. (2009), Chapter 5

Major Knowledge Representation Techniques

1. Logic-Based

- Propositional Logic
- First-Order Logic
- Description Logic

2. Structured

- Semantic Networks
- Frames
- Scripts

3. Rule-Based

- Production Rules
- If-Then Rules

4. Procedural

- Procedures
- Functions
- Programs

Key Point

Each technique has different strengths for representing different types of knowledge

Technique 1: Semantic Networks

Definition:

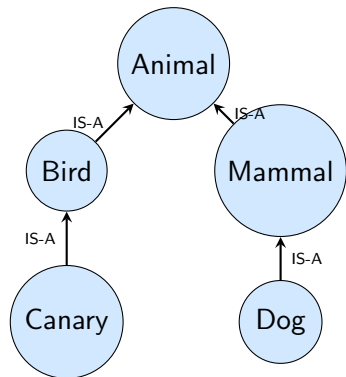
- Graph-based representation
- Nodes = concepts/objects
- Edges = relationships

Advantages:

- Intuitive and visual
- Natural for hierarchies
- Easy to navigate

Disadvantages:

- Informal semantics
- Limited expressiveness
- No standard inference



can: fly, sing

can: bark, fetch

Reference: Rich et al. (2009), Section 5.2

Technique 2: Frames

Definition: Data structures for representing stereotypical situations

Frame Example: Person

Frame: Person

Slot: Name

Value: String

Slot: Age

Value: Integer

Default: 0

Slot: Parents

Value: List of Persons

Slot: Children

Value: List of Persons

Key Features:

- **Slots:** Attributes
- **Facets:** Constraints on slots
- **Inheritance:** From parent frames
- **Defaults:** Typical values

Used for:

- Object-oriented representation
- Structured knowledge
- Expert systems

Reference: Rich et al. (2009), Section 5.3

Technique 3: Production Rules

Format: IF *condition* THEN *action*

Examples

Rule 1: IF temperature $> 100^{\circ}\text{C}$ THEN water is boiling

Rule 2: IF patient has fever AND cough THEN test for flu

Rule 3: IF stench in room THEN wumpus is nearby

Advantages:

- Modular and transparent
- Easy to add/modify rules
- Natural for experts
- Efficient forward/backward chaining

Disadvantages:

- Can become unmanageable
- Rule conflicts
- No learning mechanism
- Difficult to verify completeness

Reference: Russell & Norvig (2020), Section 9.4

Logic-Based Representations

Feature	Propositional Logic	First-Order Logic
Basic Unit	Atomic propositions	Objects, predicates, functions
Quantifiers	None	$\forall$ (for all), $\exists$ (exists)
Example	$P \Rightarrow Q$	$\forall x \text{ Human}(x) \Rightarrow \text{Mortal}(x)$
Expressiveness	Limited	High
Decidability	Decidable	Semi-decidable
Complexity	NP-complete	Undecidable in general

Trade-off

More expressive $\Rightarrow$ Higher computational complexity

Reference: Russell & Norvig (2020), Chapter 7-8

Key Issues in Knowledge Representation (1)

1. Representational Adequacy

- Can the representation express all necessary knowledge?
- Trade-off between expressiveness and efficiency

2. Inferential Adequacy

- Can new knowledge be inferred from existing knowledge?
- Are inference mechanisms sound and complete?

3. Inferential Efficiency

- How fast can inferences be made?
- Computational complexity considerations
- Trade-off: expressiveness vs. tractability

Reference: Rich et al. (2009), Section 5.1

Key Issues in Knowledge Representation (2)

4. **Acquisitional Efficiency**

- How easy is it to add new knowledge?
- Knowledge acquisition bottleneck
- Can knowledge be learned automatically?

5. **Naturalness and Transparency**

- Is representation intuitive for humans?
- Can reasoning be explained?
- Important for debugging and validation

6. **Modularity and Scalability**

- Can knowledge be organized in modules?
- Does system scale to large knowledge bases?
- Maintainability concerns

Major Challenge: The Frame Problem

Definition

The problem of representing what **doesn't change** when an action is performed.

Example Scenario

When I move from room A to room B:

- My location changes (obvious)
- But: My age doesn't change, my height doesn't change, the color of the walls doesn't change, etc.
- Need to explicitly state everything that remains the same?

Solutions:

- **Frame axioms:** Explicitly state what doesn't change
- **Effect axioms:** State only what changes
- **Successor-state axioms:** Combine both approaches

Issue: Dealing with Uncertainty

Problem: Real-world knowledge is often uncertain

Types of Uncertainty:

- 1 **Incomplete information:** Don't know everything
- 2 **Imprecise information:** Vague or approximate
- 3 **Default reasoning:** Typically true but has exceptions
- 4 **Conflicting information:** Contradictory sources

Logical Approaches:

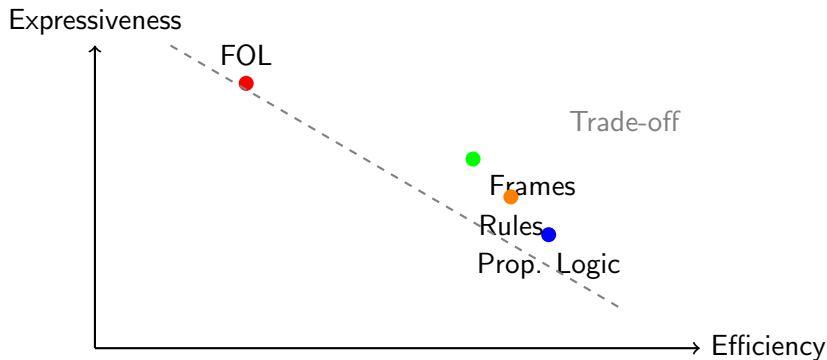
- Non-monotonic logic
- Default logic
- Modal logic

Probabilistic Approaches:

- Bayesian networks
- Fuzzy logic
- Dempster-Shafer theory

Reference: Russell & Norvig (2020), Chapter 13

Design Choices in Knowledge Representation



Key Trade-offs:

- Expressiveness $\leftrightarrow$ Computational efficiency
- Generality $\leftrightarrow$ Specialization
- Formal semantics $\leftrightarrow$ Intuitive understanding
- Automatic learning $\leftrightarrow$ Manual specification

Summary and Practical Guidance

Key Takeaways:

- Multiple KR techniques available, each with strengths/weaknesses
- Choice depends on: domain, reasoning type, efficiency needs
- Major issues: adequacy, efficiency, uncertainty, frame problem

Choosing a KR Technique:

- 1 Analyze the domain and knowledge type
- 2 Consider computational constraints
- 3 Evaluate expressiveness requirements
- 4 Assess maintainability needs
- 5 Often: hybrid approaches work best

Next Lecture

We will dive deeper into probabilistic reasoning methods to handle uncertainty

References: Russell & Norvig (2020); Rich et al. (2009)